

Here is the cleaned-up, interview-ready version. I am also correcting a few weak or wrong parts from the tutorial, because if you repeat those in an interview, it will show.

First: the mistakes in the original answers

1. **Linux vs Unix:** Linux is not just “free Unix.” More accurate: Linux is a **Unix-like** operating system kernel. Unix is older and comes from a different lineage.
 2. **Soft link behavior:** A symbolic link is **not** a copy, and writing to it does **not** “mirror” text into both places. A symlink simply points to the target file. If the target exists, you are writing to the target through the link.
 3. **vmstat -d:** That is **not** the best answer for CPU and memory stats. vmstat gives system performance stats, but for memory use free -h is clearer, and for CPU details lscpu is better.
 4. **kill:** Saying only “kill terminates a process” is too shallow. kill sends signals. kill -15 is graceful, kill -9 is forceful.
 5. **Package manager:** Good, but you should mention examples like apt, dnf, yum, pacman, and explain dependency handling.
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How to answer Linux questions in interviews

Use this pattern every time:

Definition → purpose → example command → small real-world use case

That makes your answer sound structured and not memorized.

1) What is Linux and how is it different from Unix?

Interview answer:

Linux is a **Unix-like operating system kernel** used in many distributions like Ubuntu, CentOS, Fedora, and Debian. Unix is the older family of operating systems. Linux is open-source, widely customizable, and used in servers, cloud, embedded systems, and desktops.

How to say it simply:

“Linux is open-source and Unix-like. Unix is older and usually tied to specific commercial systems.”

Example:

Ubuntu and Fedora are Linux distributions. AIX and HP-UX are Unix systems.

What not to say:

Do not say Linux is “just free Unix.” That is too vague and technically sloppy.

2) What is the Linux kernel and why is it important?

Interview answer:

The kernel is the core of Linux. It manages CPU scheduling, memory, device drivers, file systems, and networking. It acts as the bridge between software and hardware.

How to say it simply:

“The kernel is the part that controls hardware resources and decides how processes run.”

Example:

When you open Chrome, the kernel allocates memory and CPU time to it.

Command you can mention:

```
uname -r
```

This shows the kernel version.

3) What is a shell and how is it different from bash?

Interview answer:

A shell is a command interpreter that takes user commands and sends them to the OS. Bash is one specific shell. Other shells include sh, zsh, and fish.

How to say it simply:

“Shell is the category. Bash is one implementation of it.”

Example:

When you type ls, the shell interprets it and runs the command.

Command to check shell:

```
echo $SHELL
```

Interview trap:

Do not say bash and shell are the same thing. They are not.

4) What are the basic components of a Linux OS?

Interview answer:

Linux OS typically includes:

- **Kernel:** hardware management
- **Shell:** user command interface
- **System libraries:** let applications use kernel functions

- **System utilities:** tools like ls, cp, mv
- **Applications:** browsers, editors, servers

Example:

cp and mv are utilities. Firefox is an application.

5) What is the init process in Linux?

Interview answer:

init is the first process started by the kernel and has PID 1. In modern Linux, systemd usually replaces traditional init. It starts services, manages boot order, and handles system startup.

How to say it simply:

"PID 1 is the system starter. On modern Linux, that is usually systemd."

Commands:

```
ps -p 1 -o comm=  
systemctl status
```

Important:

If PID 1 dies, the system is in serious trouble.

6) How do you find files in Linux?

Interview answer:

Use find when you want to search by name, type, size, time, or permissions.

Command example:

```
find /home/user -type f -name "file1.txt"
```

More examples:

```
find . -type f -name "*.log"  
find /var/log -type f -mtime -7
```

How to say it simply:

"find searches the file system recursively and can filter by multiple conditions."

7) What is the difference between soft link and hard link?

Interview answer:

A **hard link** is another name for the same inode/data. A **soft link** or symbolic link is a pointer to a file path.

Hard link facts:

- Same inode as the original
- Still works if original filename is deleted
- Cannot cross file systems
- Usually cannot link directories

Soft link facts:

- Separate inode
- Breaks if target is deleted
- Can cross file systems
- Can point to directories

Commands:

ln original.txt hardlink.txt
ln -s original.txt softlink.txt

Important correction:

A symlink does **not** store data. It only stores a path.

8) How do you change file permissions in Linux?**Interview answer:**

Use chmod to change permissions.

Examples:

```
chmod u+x script.sh
chmod 755 script.sh
chmod 644 file.txt
```

How to explain numeric mode:

- 7 = rwx
- 6 = rw-
- 5 = r-x
- 4 = r--

How to say it simply:

“chmod controls who can read, write, or execute a file.”

9) What are the different types of permissions?

Interview answer:

Linux permissions are:

- **Read (r)**: view contents
- **Write (w)**: modify contents
- **Execute (x)**: run as a program

They apply to:

- **User/Owner**
- **Group**
- **Others**

Example:

755 means owner can read/write/execute, and others can read/execute.

Command to check:

```
ls -l
```

10) How do you create and manage symbolic links?

Interview answer:

Create a symlink with:

```
ln -s target linkname
```

Example:

```
ln -s /var/log/syslog mylog
```

How to explain it simply:

“A symlink is like a shortcut that points to the real file.”

Important correction:

Writing through a symlink changes the target file, not some copied content.

Check symlink:

```
ls -l  
readlink mylog
```

11) How do you check the current path or directory?

Interview answer:

Use pwd to print the current working directory.

Command:

pwd

Example:

If you are in /home/ashu/projects, pwd shows that exact path.

12) How do you combine commands and what is the use of a pipe?**Interview answer:**

A pipe | sends output of one command as input to another.

Example:

ls | grep "file"

Better example:

ps aux | grep python

How to say it simply:

"Pipe connects commands so I can chain filtering and processing."

Interview trap:

Do not confuse pipe with redirection. Pipe passes output to another command; redirection sends it to a file.

13) How can you check for free disk space?**Interview answer:**

Use df to see disk usage and free space on mounted file systems.

Command:

df -h

How to say it simply:

"df shows disk usage by filesystem. -h makes it human-readable."

Extra useful command:

du -sh *

This shows folder sizes inside the current directory.

14) Find .txt files containing a specific string

Interview answer:

Use grep with recursion and file filtering.

Command:

```
grep -rl "hey" --include="*.txt" /path/to/search
```

Example:

Search all .txt files under the current directory for the word error:

```
grep -rl "error" --include="*.txt" .
```

How to say it simply:

“grep searches file content, and find searches file names and metadata.”

Very important distinction:

find finds files. grep finds text inside files.

15) Ways to view a file without cat

Interview answer:

You can use:

- less for page-by-page viewing
- more for simple paging
- head for first lines
- tail for last lines
- nl for line numbers

Examples:

```
less file.txt  
head -n 20 file.txt  
tail -n 20 file.txt  
nl file.txt
```

Best interview answer:

“less is the best general alternative because it handles large files well.”

16) How do you check the current IP address of a Linux server?

Interview answer:

Use ip a or ip addr.

Command:

ip a

More specific:

ip addr show

How to say it simply:

“ip a shows network interfaces and their IP addresses.”

Old command note:

ifconfig exists on some systems but is outdated. ip is the correct modern tool.

17) What is SSH and how is it used?**Interview answer:**

SSH is a secure protocol used to connect to remote Linux machines over an encrypted channel.

Command:

ssh username@hostname_or_IP

Example:

ssh ashutosh@192.168.1.10

How to say it simply:

“SSH lets me log into another machine securely and run commands remotely.”

Useful related commands:

scp file.txt user@host:/tmp/

18) What is a package manager and why is it useful?**Interview answer:**

A package manager installs, updates, removes, and manages software along with dependencies.

Examples:

- Debian/Ubuntu: apt
- RHEL/CentOS/Fedora: dnf, yum
- Arch: pacman

Commands:

```
sudo apt update  
sudo apt install nginx  
sudo apt remove nginx
```

How to say it simply:

“It handles software installation and dependency resolution so I do not have to do it manually.”

19) How do you terminate an ongoing process?**Interview answer:**

Use kill to send a signal to a process. First try graceful termination with SIGTERM, then force it only if needed.

Commands:

```
kill PID  
kill -15 PID  
kill -9 PID  
pkill processname
```

How to say it simply:

“kill does not directly kill; it sends signals. -15 is polite, -9 is brutal.”

Process lookup:

```
ps aux | grep nginx  
top  
htop
```

Interview trap:

Do not jump to kill -9 every time. That is lazy and sometimes unsafe.

20) How do you check system architecture and CPU/memory stats?**Interview answer:**

Use lscpu for CPU architecture details. Use free -h for memory usage. Use vmstat for broader system stats.

Commands:

```
lscpu  
free -h  
vmstat 1
```

How to say it simply:

“lscpu shows CPU architecture, free shows memory, and vmstat shows system performance trends.”

Correction:

vmstat -d is about disk stats, not the best answer for CPU and memory stats.

The strongest extra Linux topics interviewers often ask

These are not optional if you want to sound serious:

File and directory basics

Know these cold:

- ls
- cd
- pwd
- mkdir
- touch
- cp
- mv
- rm

Ownership and permissions

- chmod
- chown
- chgrp
- umask
- ls -l

Process management

- ps
- top
- htop
- kill
- pkill
- jobs
- bg
- fg
- nohup

Searching and text processing

- find
- grep

awk
sed
cut
sort
uniq
tr
wc

Disk and memory

df -h
du -sh
free -h
vmstat
iostat
lscpu

Services and boot

systemctl status
systemctl start
systemctl stop
systemctl enable
journalctl

Networking

ip a
ping
curl
wget
ss -tulnp
ssh
scp
rsync

Archives and compression

tar
gzip
gunzip
zip
unzip

Scheduling

crontab -e
at

What you should say in interviews for almost any Linux question

Use this pattern:

“Linux is ...

It is used for ...

The command I use is ...

A small example is ...

One important edge case is ...”

That structure makes you sound clear and practical, not robotic.

Hard queries you should be ready for

These are tougher than the 20 basic ones:

1. Difference between **chmod** and **chown**
 2. Difference between **hard link** and **inode**
 3. Difference between **grep**, **find**, **awk**, and **sed**
 4. Difference between **kill**, **kill -9**, **pkill**, and **killall**
 5. Difference between **systemctl** and old service commands
 6. Difference between **df** and **du**
 7. What happens when a process becomes **zombie** or **orphan**
 8. What is **cron**
 9. What is **sudo** and how does it work
 10. What is the purpose of **/etc/passwd**, **/etc/shadow**, and **/etc/group**
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How to learn the hard ones properly

Do not just read definitions. That is weak preparation. Use this method:

1. **Run the command yourself**
2. **Break it intentionally**
3. **Check the output**
4. **Explain it aloud**
5. **Repeat with a real file or process**

Example for links:

```
echo hello > a.txt
ln a.txt hard.txt
ln -s a.txt soft.txt
ls -li
```

Then delete a.txt and observe what still works.

That kind of hands-on testing sticks.

Best interview-ready one-liners

Use these as compact answers:

- **Linux:** "An open-source, Unix-like OS kernel used across servers, desktops, and embedded systems."
 - **Kernel:** "The core that manages hardware and system resources."
 - **Shell:** "The command interpreter between the user and the OS."
 - **find:** "Searches files by name, type, time, size, and more."
 - **Pipe:** "Sends output of one command into another."
 - **chmod:** "Changes file permissions."
 - **ssh:** "Secure remote login to another machine."
 - **kill:** "Sends signals to processes."
 - **df:** "Shows filesystem disk usage."
 - **free:** "Shows memory usage."
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If you want, I can turn this into a **proper Linux interview cheat sheet PDF-style note** with **question** → **answer** → **command** → **example** → **trap** formatting for quick revision.